

HDI Prediction and Model Evaluation

After training the Linear Regression model, predictions are generated on the testing dataset using the **predict()** method. The predicted Human Development Index (HDI) values are stored in **y_pred** and printed for inspection.

Generate HDI Predictions: Run the trained model on **x_test** and display the predicted HDI values using **print(y_pred)**.

Calculate R-Squared (R^2): The R^2 score measures how well the independent variables explain the variation in the HDI values. A value closer to 1 indicates better predictive performance.

Test with Fewer Values: A sample input can be provided to the model to verify predictions for an individual data point.

Compare Actual and Predicted Values: Print both **y_test** (actual HDI values) and **y_pred** (predicted HDI values). The predicted values should be very close to the actual values, indicating good model performance.

```
#predicting and printing the result  
y_pred=reg.predict(x_test)  
print(y_pred)
```

```
[[0.59410218]  
 [0.93439661]  
 [0.56026001]  
 [0.69103688]  
 [0.73473957]  
 [0.50322208]  
 [0.78317645]  
 [0.92916733]  
 [0.64587963]  
 [0.66183132]  
 [0.41964029]  
 [0.91493058]  
 [0.70301803]  
 [0.94026304]  
 [0.50856461]  
 [0.66870544]  
 [0.79987143]  
 [0.7198555 ]  
 [0.72992287]  
 [0.72116564]]
```